MIL-N-52135B
15 August 1978
SUPERSEDING
MIL-N-52135A
14 December 1970

MILITARY SPECIFICATION

NETTING, CAMOUFLAGE, WIRE:

STEEL, STEEL-WOOL GARNISHED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel wire camouflage netting garnished with steel wool.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel, and Seals.
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
MMM-A-260	- Adhesive, Water-Resistant, (for Sealing Waterproofed Paper).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 1080

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-W-585	- Wire Fabric, Steel, Welded.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 141	- Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The camouflage wire netting shall consist of welded wire fabric garnished with steel wool and shall be as specified herein.

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3.2 First article (first produced camouflage wire netting). The contractor shall furnish one or more rolls of camouflage wire netting as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce camouflage wire netting that complies with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all applicable provisions of this specification (see 6.4).

3.3.1 Wire fabric. The wire fabric shall be steel, welded, conforming to MIL-W-585, Type III.

3.3.2 Garnishing. The garnishing material shall be steel wool of strands triangular in cross section and furnished in ribbons of batting 4-inches wide. Each ribbon shall contain not less than 200 and not more than 300 individual strands in any cross section. Not more than 20 percent of the strands shall measure 0.006 inch or less, over any one edge of their cross section. Ten to 30 percent of the strands shall measure 0.014 inch or larger over any one edge of their cross section.

3.4 Fabrication.

3.4.1 Fastening. The steel-wool garnishing shall be fastened to the wire fabric by wire stapling to prevent the steel-wool garnishing from shifting.

3.4.2 Coverage. The steel-wool garnishing shall be evenly distributed over the wire fabric in a uniform coverage as shown in Figure 1. The average coverage shall be not less than 12 ounces of steel-wool garnishing per square yard of wire fabric.

3.5 Treatment and painting.

3.5.1 Treatment. After assembly but prior to painting, the garnished camouflage wire netting shall be treated in accordance with TT-C-490, Type I, method as best suited.

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3.5.2 Painting. The camouflage netting shall be given two coats of paint, consisting of a baked on primer coat followed by a baked on finish coat. The paint shall completely cover all surfaces of the wire fabric and the garnishing material. All excess paint shall be removed. After thorough baking, the camouflage netting shall be allowed to cool. The paint shall not support any growth of fungus. When applied to a clear glass panel, the paint shall have a specular gloss of not more than 6.0 gloss units.

3.5.3 Weight increase after painting. After drying, the painted camouflage netting shall show a weight increase of not less than 6.7 ounces and not more than 8.2 ounces per square yard over the weight of the unpainted camouflage netting.

3.6 Color. The color of the finished camouflage netting shall be as specified (see 6.2), and shall match the color numbers shown in Table I, which conform to FED. STD. No. 595.

3.7 Infrared reflectance. The infrared reflectance of the finished camouflage netting shall be within the applicable limits specified in Table I both before and after being subjected to the accelerated weathering test.

TABLE I. Infrared reflectance of colored netting.

Color	Color chip number of FED. STD. No. 595	Infrared reflectance Percent minimum	Percent maximum
White	37875	57.0	100.0
Black	37038	0.0	24.5
Earth brown	30099	24.5	57.0
Field drab	30118	24.5	57.0
Earth yellow	30257	24.5	100.0
Sand	30277	24.5	100.0
Earth red	30117	24.5	57.0
Desert sand	30324	24.5	100.0
Forest green	34079	24.5	57.0
Dark green	34102	37.0	57.0
Olive drab	34087	24.5	57.0
Light green	34127	37.0	57.0

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3.8 Dimensions. The finished camouflage wire netting shall be furnished in rolls 72-inches wide plus or minus 0.50 inch. Unless otherwise specified (see 6.2), the length of netting in each roll shall be 75 feet plus or minus 1.50 feet.

3.8.1 Splicing. Sections of the camouflage wire netting requiring splicing to form a roll shall be not less than 10 feet in length. A roll shall contain not more than two splices. The splicing shall be done neatly and care shall be taken to leave no protruding wire ends that may cause injury to personnel. Not more than 10 percent of the rolls in any lot shall contain splices.

3.9 Flame resistance. When subjected to a flame from a Bunsen burner, the painted steel wool shall show no spreading of flame.

3.10 Corrosion-resistance. After 100 hours exposure to salt spray, the camouflage wire netting shall show no evidence of corrosion.

3.11 Weathering resistance. After 300 hours of accelerated weathering, the camouflage wire netting shall show no appreciable fading, darkening, or chalking. Slight changes in the lightness value and saturation of a color are permissible, but the hue or color purity shall remain constant. For example, a yellow-green color may be lighter or darker (more or less vivid and brilliant) by reason of grayness, but shall not become a blue-green.

3.12 Identification marking. The finished camouflage wire netting shall be identified in accordance with MIL-STD-130.

3.13 Workmanship. The finished camouflage wire netting shall be free from defects such as rust, breaks, protruding wire ends, or other defects that might impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced camouflage wire netting inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced camouflage wire netting inspection.

4.3.1 Examination. The first produced camouflage wire netting shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced camouflage wire netting shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed roll of camouflage wire netting shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects listed in 4.5.1. AQL shall be 6.5 percent defective for major defects and 10.0 percent defective for minor defects.

4.4.4 Tests.

4.4.4.1 Paint samples. Paint which is required to be used in tests shall be random sampled from paint being used in production. These samples shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3. Failure of any test shall be cause for rejection of the batch of paint represented by the samples.

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4.4.4.2 Camouflage wire netting samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.4, 4.5.2.5, and 4.5.2.6. AQL shall be 6.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The camouflage wire netting shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Fabrication not as specified.
103. Treatment and painting not as specified.
104. Color not as specified.
105. Dimensions not as specified.
106. Splicing not as specified.
107. Workmanship not as specified.

Minor

201. Identification marking improper, illegible, or missing.

4.5.2 Tests.

4.5.2.1 Fungus resistance (paint only). Determine the fungus resistance of the paint in accordance with FED. TEST METHOD STD. No. 141, Method 6271.1, except that:

(1) Test organism used shall be Aspergillus oryzae, ATCC No. 10196, or QM 1273.
(2) Test specimens shall be leached before inoculation.
(3) The sterile distilled water used in preparing the inoculum shall contain approximately 0.005 percent of a nontoxic wetting agent.
(4) The criteria for acceptance shall be total absence of fungus growth on the specimens after the specified period of incubation.

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4.5.2.2 Specular gloss (paint only). Determine the specular gloss unit of the paint in accordance with FED, TEST METHOD STD. No. 141, Method 6101. Test specimen shall be a sample of paint drawn on a clear glass panel. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3 Infrared reflectance (paint only). Measure the infrared reflectance value of the paint in accordance with FED. TEST METHOD STD. No. 141, Method 6241.1. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.3.1 Preparation of test specimen. A 3- by 5-inch section of the finished camouflage wire netting shall be placed on a 3- by 5-inch smooth-surface paper background. The specimen may be fastened to the paper background by mechanical means; adhesives shall not be permitted. The paper background shall have the following characteristics:

(a) Line-divided into two equal panels, one panel shall be white and the other black. The black panel shall have a diffuse reflectance of not less than 75 percent.
(b) The coated paper shall be impervious and resistant to the paint used on the camouflage wire netting.

4.5.2.4 Flame resistance. Obtain a portion of the coated steel-wool garnishing from the sample roll of finished camouflage wire netting and draw into a skein, cylindrical in form, having a diameter of approximately 1/2 inch and a length of approximately 9 inches, weighing approximately 1/2 gram per inch. With the skein in a horizontal position, test the flame resistance of the skein by means of a Bunsen burner using a nonluminous flame approximately 3 inches in length with the tip of the interior cone of the flame just reaching the lower end of the skein. Hold the flame in this position for 10 seconds and examine the burn. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.5 Salt spray. Expose a 3- by 6-inch specimen of finished camouflage wire netting, obtained from the sample roll, to 100 hours of salt spray in accordance with FED. TEST METHOD STD. No. 141, Method 6061. Nonconformance to 3.10 shall constitute failure of this test.

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4.5.2.6 Accelerated weathering. Expose two test specimens of finished camouflage wire netting, obtained from the sample roll, to 300 hours of accelerated weathering in accordance with FED. TEST METHOD STD. No. 141, Method 6151 or 6152. Nonconformance to 3.11 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials not as specified for Level A.
109.	Sealing of materials not as specified for Level A.
110.	Girthwise bands of tape not applied as specified for Level A.
111.	Boxes not as specified for Level A or Level B.
112.	Quantities packed together exceed limitations of the box for Level A or Level B.
113.	Strapping not as specified for Level A.
114.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each roll of netting shall be wrapped with barrier material conforming to PPP-B-1055, Class E-1 with all folds, seams, and laps sealed with adhesive conforming to MMM-A-260 or with tape conforming to PPP-T-60, Type IV. As an alternate, each roll of netting

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may be wrapped with polyethylene conforming to L-P-378, Type II, Class 1, grade optional, Finish 1, film thickness 0.0060 inch. Sealing of all folds, laps, and seams shall be by heat sealing or with tape specified herein. Each roll, wrapped in barrier material or polyethylene, shall then be secured with three bands of tape conforming to PPP-T-97, Type IV, Size 1/2 inch, applied girthwise completely around the roll, and the tape overlapped on itself a minimum of 8 inches. Two bands shall be positioned approximately 18 inches from each end and the other approximately in the center.

5.1.2 Commercial. Each roll of netting shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Rolls of netting, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Rolls of netting, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-640, Class 2, style optional or PPP-B-636, V3c, V11c, V13c, or V15c, style optional, may be used when the quantity to be packed does not exceed the weight or size limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Rolls of netting, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The steel-wool garnished camouflage wire netting is intended for use in construction pertaining to concealment of installations and activities from enemy observations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced camouflage wire netting and number of rolls required (see 3.2).
(c) Color required (see 3.6).
(d) Length of roll if other than as specified (see 3.8).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(f) When other than Finish B strapping is required (see 5.2.1).

6.3 First produced camouflage wire netting. Any changes or deviations of production camouflage wire nettings from the first produced camouflage wire netting during production will be subject to the approval of the contracting officer. Approval of the first produced camouflage wire netting will not relieve the contractor of his obligation to furnish camouflage wire nettings conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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FIGURE 1. Steel-wool garnished netting

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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